STALKER

Stock Market Analyzer & Alpha Engine v3.0

Complete System Documentation

Data Analysis Pipeline | Scoring Models | Risk Management

Last Updated: June 3, 2026

STALKER ? System Architecture & Complete Screening Process

A Technical and Quantitative Guide to the STALKER Algorithmic Stock Screener

---

1. System Overview

STALKER is an automated, institutional-grade quantitative screening and portfolio construction system designed for the Indian stock market (NSE & BSE). The system executes a daily multi-stage pipeline to process a universe of 188 liquid symbols, analyze market context, rank stocks based on a hybrid Alpha-Expected Value model, validate them through a Leadership layer, perform risk management, and assemble a correlation-optimized portfolio.

```
 graph TD
A[188 Stock Universe] --> B[Stage 1: Hard Safety Filters]
B --> C[Stage 2: Regime & Pulse Engines]
C --> D[Stage 3: Ensemble Alpha Engine & EV Model]
D --> E[Stage 4: Leadership Validation Layer]
E --> F[Stage 5: Reality Check & Dual Gating]
F --> G[Stage 6: Drawdown-Aware Sizing]
G --> H[Stage 7: Portfolio Assembly]
H --> I[E-Mail Dispatch & DB Log]
```

---

2. Daily Execution Workflow

The STALKER scheduler operates on a strict daily timeline designed around the Indian market sessions (IST). Each step corresponds to a specific module or execution mode in main.py.

Time (IST)	Module / Script	Action
7:00 AM	main.py --mode run	Morning Scan: Executes screener.run_screen(). Perf
8:15 AM	main.py	Price Verification: Cross-checks entry prices agai
8:30 AM	main.py	Morning Email Dispatch: Dispatches the morning ema
9:20 AM	main.py	Opening Print Recording: Records the actual openin
3:35 PM	main.py	Closing Print Recording: Records the daily closing
4:00 PM	generate_mistakes_audit.py	EOD Report & Audit: Calculates EOD P&L, audits set

---

3. Step-by-Step Screening Pipeline

Step 1 ? Data Fetching & Prep Pipeline

Before checks run, the data fetcher (data_fetcher.py) performs a two-pass chunked parallel download of historical data using yfinance to avoid rate limits and memory leaks:

- Pass 1 (3-Month Daily Data): Loaded for all universe stocks to calculate short-term technicals, ATR, and VCP.
- Pass 2 (1-Year Daily Data): Loaded selectively for stocks that pass initial filters to evaluate long-term moving averages, Minervini trend template conditions, and trend stability scores.

Step 2 ? Stage 1: Hard Safety Filters (The Gatekeepers)

Stocks must pass every single gate below in screener.py to become candidates:

1. Liquidity Gate (evaluate_liquidity):

- Computes the 20-day average volume and daily traded value:

$$\text{Daily Turnover} = \text{Average Volume (20d)} \times \text{Latest Close}$$

- Threshold: Must exceed config.MIN_DAILY_TURNOVER (?10 Crore/day or 100,000,000 INR).

2. Overnight Gap-Risk Filter (check_overnight_gap_risk):

- Evaluates historical gap sizes over the last 20 trading days:

$$\text{Gap}\% = \left|\frac{\text{Open}_t - \text{Close}_{t-1}}{\text{Close}_{t-1}}\right| \times 100$$

- Rejection: Fails if $\text{Gap}\% > 3.0\%$ on more than 5 of the past 20 trading days.

3. ATR Volatility Spike Filter (check_atr_spike_risk):

- Rejection: Rejects any stock whose latest ATR(14) is $> 2.0\times$ its 20-day average ATR. This filters out stocks undergoing erratic, non-structural volatility shocks.

4. Circuit-Lock Filter (check_circuit_lock):

- Rejection: Rejects if High == Low (locked in circuit) or daily volume is < 5000 shares and $< 5\%$ of its 20-day average volume.

5. Calibrated Risk Gating (evaluate_risk_profile):

- Computes a standardized 0?10 risk score combining ATR volatility and 60-day maximum drawdown:

$$\text{Risk Score} = (\text{ATR}\% \times 150) + (\text{Max Drawdown}\% \times 15)$$

where:

- $\text{ATR}\% = \frac{\text{Latest ATR}(14)}{\text{Latest Close}}$
- $\text{Max Drawdown}\% = \text{Max drawdown over the last 60 trading days}$
- Dynamic Gating Threshold:
- Bear / Deteriorating Regime: Max allowed Risk Score = 5.2

- Neutral / Sideways / Improving Regime: Max allowed Risk Score = 6.2
- Bull Regime: Max allowed Risk Score = 7.2

6. Data Quality Gate (calculate_data_quality):

- Scores data freshness and completeness. Rejects if score is < 70.0 or if critical fundamental fields (ROE, Debt/Equity, Revenue Growth) are missing.

7. Downtrend Rejection (detect_market_structure):

- Runs swing detection and moving average structures. Instantly rejects if the stock is in a confirmed downtrend.

Step 3 ? Stage 2: Regime Classifier & Market Pulse

1. Regime Engine (regime_engine.py):

- Classifies the market into one of 8 states using Nifty 50 position relative to EMA50 and EMA200, Nifty 20d annualized realized volatility, EMA20 slope, and universe market breadth (% stocks above 50 and 200 EMA):
- Bull_Trend: Nifty Close > EMA50 and Nifty Close > EMA200 (normal volatility and breadth).
- Bull_Expansion: Close > EMA50 and Close > EMA200, Market Breadth (50 EMA) >= 60%, Breadth rising, and EMA20 slope > 0.05.
- Bull_Exhaustion: Close > EMA50 and Close > EMA200, Realized Volatility > 22.0%, and Market Breadth (50 EMA) < 40%.
- Bear_Trend: Close < EMA200 (normal conditions).
- Bear_Panic: Close < EMA200, Realized Volatility > 35.0%, and Advance/Decline ratio < 0.5 (capitulation).
- Bear_Recovery: Close < EMA200, Market Breadth (50 EMA) rising, Advance/Decline ratio > 1.2, and EMA20 slope > 0.0.
- Neutral_Compression: Between EMA50 and EMA200, Realized Volatility <= 22.0%, |EMA20 slope| < 0.05, and breadth not rising.
- Neutral_Rotation: Between EMA50 and EMA200 (not compression).
- Buying Permission: Buys are suppressed entirely if the regime is Bear_Trend or Bear_Panic.

2. Market Pulse Engine (market_pulse.py):

- Computes a daily buyer-vs-seller pressure metric (0?100) combining India VIX, universe Advance/Decline ratio, closing range positioning, and Chaikin Money Flow.
- Pulse Gate: If Pulse Score is < 35, all BUY picks are automatically downgraded to WATCH due to extreme selling pressure.

3. Sector RS Momentum:

- Computes relative performance of sectors vs Nifty over a 20-day window:

$$\text{Sector RS} = \text{Sector 20d Return}\% - \text{Nifty 50 20d Return}\%$$

- Stage 1 survivors in the top 25% highest-performing sectors and industries are flagged as leading_sectors and leading_industries.

Step 4 ? Stage 3: Ensemble Alpha Engine & EV Model

1. Alpha Scoring Sub-Models:

- Momentum Sub-Model: Uses relative strength percentile vs Nifty, EMA alignment, and RSI positioning (using a Gaussian curve centered at 60).
- Quality Sub-Model: Uses capital efficiency (ROE) and growth stability.
- Institutional Sub-Model: Uses volume ratio surges (relative to 20-day average) and Chaikin Money Flow.
- Catalyst Sub-Model: Uses analyst earnings surprises and EPS growth acceleration.

2. Ensemble Weights Matrix:

- The sub-model scores are combined using weights dynamically adapted to the 8 market regimes:

Regime	Momentum	Quality	Institutional	Catalyst
Bull_Trend	40%	20%	25%	15%
Bull_Exp	45%	15%	25%	15%
Bull_Exhaustion	20%	35%	30%	15%
Neutral_	25%	30%	25%	20%
Neutral_Compression	20%	35%	25%	20%
Bear_Tr	10%	45%	30%	15%
Bear_Panic	05%	50%	35%	10%
Bear_Re	25%	35%	25%	15%

3. Meta Model Adjustment (meta_model.py):

- Classifies setup/trade types into BREAKOUT, PULLBACK, MOMENTUM, VALUE_MOMENTUM, or EARNINGS_RUNNER and adjusts raw alpha scores.

4. Expected Value (EV) Hybrid Model:

- Blends the adjusted Alpha score with setup Expectancy:

$$\text{Expectancy} = (\text{Win Rate} \times \text{Avg Win}) - (\text{Loss Rate} \times \text{Avg Loss})$$

$$\text{EV Score} = \min(100.0, \max(0.0, 50.0 + \text{Expectancy} \times 10))$$

$$\text{Final Score} = \text{Alpha Weight} \times \text{Meta Alpha} + \text{EV Weight} \times \text{EV Score}$$

where:

- $\text{EV Weight} = 0.30 \times \text{EV Confidence}$
- $\text{EV Confidence} = \min(1.0, \frac{\text{Sample Size}}{30})$
- $\text{Alpha Weight} = 1.0 - \text{EV Weight}$

5. Regime Penalties:

- Non-defensive sectors (e.g. IT, Banking) receive a flat -15.0 point penalty in Bear legacy markets.
- Capped total penalty of -30.0 points for debt/equity > 1.2 (-10), ATR% > 6% (-10), and missing earnings (-5).

6. Momentum Leadership Boost:

- Adds a +7.5 point boost if the stock has RS rank $\ge 75\%$ and belongs to a leading sector or leading industry.

Step 5 ? Stage 4: Leadership Validation Layer (Elite Qualification)

Candidates are validated through leadership_engine.py to identify true institutional leading structures:

1. Mark Minervini Trend Template (8 Conditions):

- Condition 1: Latest Close > 150-day SMA and Close > 200-day SMA (Non-negotiable).
- Condition 2: 150-day SMA > 200-day SMA (Non-negotiable).
- Condition 3: 200-day SMA is trending up for at least 1 month (22 trading days).
- Condition 4: 50-day SMA > 150-day SMA and 50-day SMA > 200-day SMA.
- Condition 5: Latest Close > 50-day SMA (Non-negotiable).
- Condition 6: Close >= 1.30 * 52-week Low (Non-negotiable).
- Condition 7: Close >= 0.75 * 52-week High (within 25% of 52w high).
- Condition 8: Relative Strength (RS) Rank >= 70.0.
- Rejection Gate: Rejects if conditions passed is less than config.MINERVINI_MIN_CONDITIONS (relaxed to 5/8) or if any non-negotiable condition fails.

2. Volatility Contraction Pattern (VCP) Detection:

- Programmatically detects consolidations: contracts (2?4 troughs), shrinking pullback depth, ATR contraction (ATR 10 < ATR 20), tight closes, and volume tapering.
- VCP Scoring (0-100):
- Price close to 60-day high (within 12%): +20 points.
- ATR Compression (ATR 10 < ATR 20): +20 points.
- Tight Closes (max spread < 2.5% over 5 days): +15 points.
- Volume Tapering (10d avg volume < prior 20d avg): +15 points.
- Contractions count: 2 contractions (+15 points), 3+ contractions (+30 points).
- Shrinking contraction depth: +10 points.
- Grades: Elite (Score >= 80), Strong (60-79), Weak (40-59), None (< 40). Fails validation if VCP score is < 40.0.

3. Leadership Stability Score:

- Measures the percentage of the last 60 days that the stock has traded in a healthy structure: Close > 150 SMA and 50 SMA > 150 SMA.

4. Leadership Score:

- Unified score:

$$\text{Leadership Score} = (\text{Stability Score} \times 0.35) + (\text{Sector RS} \times 0.25) + (\text{Industry RS} \times 0.20) + (\text{Market Bull Bonus} \times 0.10) + (\text{Inst Score} \times 0.10)$$

where Market Bull Bonus = 100 if Nifty is bullish, else 0.

5. Final Multipliers:

- Gearing applied to Alpha rankings:

$$\text{Final Score} = \text{Alpha Score} \times \text{Leadership Multiplier} \times \text{VCP Multiplier}$$

- VCP Multipliers: Elite (1.07x), Strong (1.04x), Weak (1.02x), None (1.00x).
- Leadership Multipliers: Elite (1.05x), Strong (1.02x), Acceptable (1.00x), Low (0.95x).

Step 6 ? Stage 5: Reality Check & Dual Gating

1. Friction Checks (reality_check.py):

- Downgrades BUY to WATCH if:
- Gap Extension: Opening gap exceeds 5%.
- Earnings Proximity: Earnings event within 3 days.
- Circuit Proximity: Close within 1% of 52-week high (circuit proxy).
- Minimum Spread: Average daily price range over last 10 days is $< 0.15\%$ of Close.

2. Dual-Filter Score & Percentile Gates:

- Candidates must pass dynamic score and percentile gates based on N (number of candidates):
- Bull Regime: $\text{Score} \ge 65.0$ and $\text{Rank} \le \max(5, 25\%\text{ of } N)$
- Neutral / Improving: $\text{Score} \ge 65.0$ and $\text{Rank} \le \max(3, 15\%\text{ of } N)$
- Bear / Deteriorating: $\text{Score} \ge 70.0$ and $\text{Rank} \le \max(2, 10\%\text{ of } N)$

3. Risk Management & R:R Checks:

- Stop loss is calculated as the wider of the 2.0x ATR stop and 1% below the recent swing support (capped at a maximum of 10% loss):

$$\text{Stop Loss} = \max\left(\min(\text{Latest Close} - 2.0 \times \text{ATR}(14), \text{Swing Support} \times 0.99), \text{Latest Close} \times 0.90\right)$$

- Targets are set at:
- Target 1 = Entry + 1.5 * Risk (Conservative 1:1.5)
- Target 2 = Entry + 2.0 * Risk (Ideal 1:2)
- Target 3 = Entry + 3.0 * Risk (Stretch 1:3)
- Rejects BUY action (downgrades to WATCH) if risk-to-reward ratio is < 1.5.

Step 7 ? Stage 6: Drawdown-Aware Sizing

Saves capital by reducing position sizing dynamically based on historical account drawdowns (risk_manager.py):

Account Equity Drawdown	Position Sizing Multiplier	Action / Sizing Description
0% to 5.0%	1.0x	Full Sizing (2% capital risk per trade)
5.0% to 10.0%	0.5x	Half Sizing (1% capital risk per trade)

10.0% to 15.0%	0.25x	Quarter Sizing (0.5% capital risk per trade)
> 15.0%	0.0x	Trading Halted (Protect Capital)

Step 8 ? Stage 7: Portfolio Assembly & Correlation

Calculates absolute pairwise return correlations of candidate stocks over 20-day and 60-day windows (portfolio_engine.py):

1. Sector Exposure Limit: Maximum of 2 stocks from the same sector.
2. Industry Exposure Limit: Maximum of 1 stock from the same industry.
3. Correlation Gate:

- Average absolute pairwise return correlation across all candidates must be ≤ 0.60.
- Single-stock absolute 60-day return correlation must be ≤ 0.80 against each existing portfolio position.

4. Total Exposure (Heat Limit): Caps portfolio risk exposure (active heat + new entries) at 6% of capital.
5. Market Pulse Gate: Overrides all BUY choices to WATCH if the buyer pressure pulse score is < 35.
6. System Kill Switch: Overrides all BUY choices to WATCH if consecutive loss limits are breached.

4. Parameter Configurations Matrix (config.py)

The following configurations govern the thresholds and behaviors of the screening pipeline:

Parameter	Type	Value	Scope / Description
MIN_DAILY_TURNOVER	Integer	100,000,000	Stage 1: Liquidity threshold (?10 Crore/day)
MAX_STC	Integer	10,000	Price filter: Max stock price in INR
MIN_STOCK_PRICE	Integer	50	Price filter: Min stock price to avoid penny stock
STRICT_E	Boolean	False	Gating: Allows top leadership buys in Neutral/Impr
MINERVINI_MIN_CONDITIONS	Integer	5	Stage 4: Requires 5/8 passed Mark Minervini templa
VCP_MIN_	Float	40.0	Stage 4: Quality score baseline for VCP pattern va
PORTFOLIO_MAX_RISK_PCT	Float	0.06	Stage 7: Max 6% total capital exposure (heat limit
RISK_PEF	Float	0.02	Sizing: Max 2% capital risk per trade at 1.0x sizi
STOP_LOSS_ATR_MULT	Float	2.0	Sizing: Multiplier for ATR-based stop loss distanc
DAILY_LO	Float	0.03	Safety: Hhalts trading if daily drawdown reaches 3
KILL_SWITCH_LOSS_COUNT	Integer	5	Safety: Breaching 5 consecutive losses triggers Ki
KILL_SWI	Integer	3	Safety: Suspend trading for 3 days on Kill Switch
MIN_RISK_REWARD	Float	1.5	Sizing: Minimum risk-to-reward ratio for trade ent
MAX_DEE	Float	1.5	Fundamental: Max debt-to-equity ratio threshold

MIN_PROMOTER_HOLDING	Integer	35	Fundamental: Minimum promoter holding percentage
MIN_MAR	Integer	5,000	Fundamental: Minimum market cap in crores (?5,000